

Andrea Maselli

Curriculum Vitae

Gran Sasso Science Institute,
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Personal Information

Full Name **Andrea Maselli**
Place of Birth **Tricarico (IT)**
Date of Birth **March 12th, 1985**
Nationality **Italian**

Present Position

Dec 2023–on **Associate Professor in Theoretical Astrophysics**, *Gran Sasso Science Institute (GSSI)*, L'Aquila (IT).

Past Positions

Sep 2025– **Fellow of the Italian Academy**, Columbia University, New York (USA).
Dec 2025
Dec 2020– **RTD-B Tenure-Track Assistant Professor in Theoretical Astrophysics**, *Gran Sasso Science Institute (GSSI)*, L'Aquila (IT).
Nov 2023
Oct 2018– **RTD-A Assistant Professor**, *Department of Physics*, Sapienza University of Rome, Rome (IT).
Nov 2020
Oct 2017– **Postdoctoral Researcher**, *CENTRA – Multidisciplinary Center for Astrophysics*, Instituto Superior Técnico, Lisbon (PT).
Sep 2018
Oct 2015– **Postdoctoral Researcher**, *Institute of Astronomy and Astrophysics*, Eberhard Karls University of Tübingen, Tübingen (DE).
Sep 2017
Jan 2015– **Postdoctoral Researcher**, *Center for Relativistic Astrophysics*, Georgia Institute of Technology, Atlanta, GA (USA).
Jun 2015
May 2014– **Postdoctoral Researcher**, *Department of Physics*, Sapienza University of Rome, Rome (IT).
Dec 2014
May 2013– **VESF Postdoctoral Fellow**, *Department of Physics*, Sapienza University of Rome, Rome (IT).
Apr 2014
Nov 2012– **Postdoctoral Fellow**, *Institute of Cosmology and Gravitation*, University of Portsmouth, Portsmouth (UK).
Apr 2013

Education

- 2025 **National Scientific Habilitation (ASN)**, Full Professor in Astronomy, Astrophysics, Earth and Planetary Physics, Sector 02/C1.
- 2025 **National Scientific Habilitation (ASN)**, Full Professor in Theoretical Physics of Fundamental Interactions, Sector 02/A2.
- 2020 **National Scientific Habilitation (ASN)**, Associate Professor in Astronomy, Astrophysics, Earth and Planetary Physics, Sector 02/C1.
- 2018 **National Scientific Habilitation (ASN)**, Associate Professor in Theoretical Physics of Fundamental Interactions, Sector 02/A2.
- 2009–2013 **PhD in Theoretical Physics**, *Sapienza University of Rome*, Thesis: "Tidal Interactions in Compact Binary Systems".
Advisor: Prof. V. Ferrari
Main topics: tidal interactions in black hole and neutron star systems; gravitational-wave emission from relativistic sources; post-Newtonian approximation. Defended January 24, 2013.
- 2006–2009 **MSc in Theoretical Physics**, *Sapienza University of Rome*, 110/110 *cum laude*; Thesis: "Monte Carlo Markov Chains in Cosmology".
Advisor: Prof. A. Melchiorri
Main topics: stochastic methods for multi-parameter data analysis; cosmic microwave background radiation; primordial nucleosynthesis. Defended October 10, 2009.
- 2003–2006 **BSc in Physics and Astrophysics**, *Sapienza University of Rome*, 110/110; Thesis: "CMB Anisotropies and Polarization".
Advisor: Prof. A. Melchiorri
Main topics: cosmic microwave background radiation; characterization of temperature and polarization anisotropies. Defended December 14, 2006.

Awards, Honors & Fellowships

- 2026–2028 **Junior Invited Professor (fellowship)**, Space Academy – Science and New Space Chair, Paris Observatory, for the project "Waveform Modelling of Binary Black Hole Mergers in Astrophysical Environments".
Three 1-month research stays (extendable to six months total; funded visiting position)
- 2025 **Fellowship**, 39K\$, Italian Academy for Advanced Studies in America, Columbia University, for the project "Binary Black Hole Systems as Particle Physics Laboratories in Dark Matter Environments".
- 2019 **Honorable Mention**, Gravity Research Foundation Essay Competition, for "*Love in Extrema Ratio*" ([link](#)).
- 2018 **RTDA Postdoctoral Fellowship** (Associate Researcher), for the project *Theory and Phenomenology of Gravitational Waves*, Amaldi Research Center, Sapienza University of Rome.
- 2014 **Best Contribution by a Young Scientist**, 3rd Forum on Gravitational Wave Astrophysics, Sant Cugat Forum on Astrophysics, Sant Cugat (Spain).
- 2013–2014 **VESF Postdoctoral Fellowship**, 25K€, European Gravitational Observatory (EGO), for the project "Tidal Interactions in Compact Binaries and the Equation of State of Nuclear Matter".
- 2012–2013 **Fellowship for Scientific Research Abroad**, 7.8K€, Sapienza University of Rome, for the project "Relativistic Compact Binary Sources and Gravitational Waves".

Research Funding

For collaborative grants, reported amounts refer to both the share awarded to my unit and the total project funding.

- 2026 **Horizon Europe MSCA Staff Exchanges 2025**, *PI*, 466K€, European Union, for the project "Strong Gravity Crossroads: Fundamental Fields in the High-Energy Universe" (STRONG).
- 2025 **Workshop**, "*Crossroads in Strong Gravity: Challenges & Future Directions*", 15K€, Gravity Theory Trust.
- 2024 **Royal Society International Exchange**, *Co-I (unit lead)*, 7.25K€ [14.5K€ total], Royal Society, for the project "Gravitational Wave Observations of Asymmetric Binaries as Probes of Fundamental Fields".
- 2024 **Workshop**, "*Crossroads in Strong Gravity: Challenges & Future Directions*", 4K€, INFN.
- 2023 **PRIN 2022**, *Co-I (unit lead)*, 87.5K€ [175K€ total], Italian Ministry of University and Research, for the project "Gravitational Wave Observations of Asymmetric Binaries as Probes of Fundamental Fields".
- 2023 **Italy–USA Bilateral Agreement**, *PI*, 105K€, Italian Ministry of Foreign Affairs, for the project "Fundamental Physics and Astrophysics with Next-Generation Gravitational Wave Detectors".
- 2021 **Short-Term Scientific Mission**, 1.8K€, COST Action GWverse (CA16104), Research visit at Instituto Superior Técnico, Lisbon (PT).
- 2020 **Sapienza Research Grant**, *PI*, 13K€, Sapienza University of Rome, for the project "Exploring Fundamental Physics with Gravitational Waves".
- 2019 **Workshop**, "*Lost in Gravity*", 8K€, COST Action GWverse (CA16104). Organized with M. Zilhão and L. Bernard
- 2019 **Short-Term Scientific Mission**, 1.4K€, COST Action GWverse (CA16104), Research visit at University of Nottingham, Nottingham (UK).
- 2017 **Workshop**, "*Neutron Stars in Lisbon*", 5K€, COST Action PHAROS (CA16214).
- 2017 **Short-Term Scientific Mission**, 1K€, COST Action PHAROS (CA16214), Research visit at University of Tübingen, Tübingen (DE).

Participation in collaborative projects

- 2022–2025 **PRIN 2021**, *Multimessenger Astronomy in the Einstein Telescope Era (METE)*, Italian Ministry of University and Research, 577K€, National Coordinator: M. Branchesi; Local Coordinator: M. Branchesi.
Team Member
- 2021–2025 **MSCA RISE (H2020)**, "*Gravitational Universe: Challenges and Opportunities*", EU Horizon 2020, 280K€, Global Coordinator: L. Gualtieri.
Team Member
- 2019 **PRIN 2017**, *Black Holes in "The Gravitational Universe"*, Italian Ministry of University and Research, 211K€, National Coordinator: M. Colpi; Local Coordinator: P. Pani.
Team Member
- 2016–2019 **MSCA RISE (H2020)**, "*Strong Gravity and High-Energy Physics*", EU Horizon 2020, 288K€, Global Coordinator: U. Sperhake.
Team Member

2012–2015 **FP7-IRSES**, "*Numerical Relativity and High-Energy Physics*", European Commission, 160K€,
Global Coordinator: C. Herdeiro.
Team Member

Experience

Major research collaborations and membership of scientific societies

2026–on **INAF Associate**, *Istituto Nazionale di AstroFisica (INAF)*.
2022–on Member of **Einstein Telescope Observational Science Board**.
2022–on Member of **Lunar Gravitational Wave Antenna Science Team**.
2018–on Member of **Portuguese Society of Relativity and Gravitation**.
2018–on **INFN Associate**, *Istituto Nazionale di Fisica Nucleare (INFN)*.
2021–2025 Member of **Italian Society of General Relativity and Gravitation**.
2019–2025 Member of **International Society on General Relativity and Gravitation**.
2018–2025 Member of **LISA Consortium**.
2017 Member of **eXTP Science Team**.
2018–2021 Member of **3G Science Team**.

Coordination activity within major research collaborations

2022–on **Local Coordinator at GSSI for the INFN national initiative TEONGRAV**, *Theory of Gravitational Wave Sources*.
2022–2026 **Coordinator of Einstein Telescope OSB Division 9**, *Scientific potential of detector configurations and common tools*.
2022–on **Coordinator of Working Group 1, Lunar Gravitational Wave Antenna**, *Gravitational-wave science and multi-messenger astronomy*.
2024–2025 **Local Coordinator at GSSI of the LISA Consortium group**.

Lectures of university courses

Mar 2026–on Lecturer for "**The Dynamics of Asymmetric Binaries**" [PhD course], *Gran Sasso Science Institute, L'Aquila (IT)*.
Oct 2026 Lecturers "**Gravitational Waves**" and "**Neutron Stars and their Astrophysics**" [Master course], *Meudon, Paris (FR)*.
Mar 2024– Mar 2025 Lecturer for "**Black Hole Perturbation Theory**" [PhD course], *Gran Sasso Science Institute, L'Aquila (IT)*.
Nov 2021– Oct 2024 Lecturer for "**Gravitation and Cosmology**" [PhD course], *Gran Sasso Science Institute, L'Aquila (IT)*.
Course taught with Prof. S. Matarrese
Mar 2023 Lecturer for "**The Physics of Black Holes**" [PhD course], *Gran Sasso Science Institute, L'Aquila (IT)*.
Course taught with Prof. T. Sotiriou
Mar 2022 Lecturer for "**Physics of Compact Stars**" [PhD course], *Gran Sasso Science Institute, L'Aquila (IT)*.
Course taught with Prof. M. Mannarelli
Mar 2020– Oct 2020 Lecturer for "**Relativistic Perturbations of Compact Objects**" [PhD course], *Sapienza University of Rome, Rome (IT)*.

- Mar 2019– Nov 2020 Lecturer for **"Complex Analysis" [undergraduate course]**, *Sapienza University of Rome*, Rome (IT).
Course taught with Prof. P. Santini
- Apr 2017– Jul 2017 Lecturer for **"Black Hole Physics" [graduate course]**, *Eberhard Karls University of Tübingen*, Tübingen (DE).
- Apr 2017– Jul 2017 Teaching Assistant for **"Astronomy and Astrophysics" [undergraduate course]**, *Eberhard Karls University of Tübingen*, Tübingen (DE).
Course taught by Prof. A. Santangelo
- Mar 2016– Jul 2017 Teaching Assistant for **Experimental Applied Physics [undergraduate course]**, *Eberhard Karls University of Tübingen*, Tübingen (DE).
Course directed by Prof. T. Hehl
- Oct 2015– Feb 2017 Teaching Assistant for **"Introduction to General Relativity" [graduate course]**, *Eberhard Karls University of Tübingen*, Tübingen (DE).
Course taught by Prof. K. Kokkotas
- Apr 2016– Jul 2016 Teaching Assistant for **"Astronomy and Astrophysics" [undergraduate course]**, *Eberhard Karls University of Tübingen*, Tübingen (DE).
Course taught by Prof. W. Kley
- Oct 2015– Feb 2016 Teaching Assistant for **"Numerical Methods for Physics and Astrophysics" [undergraduate course]**, *Eberhard Karls University of Tübingen*, Tübingen (DE).
Course taught by Prof. K. Kokkotas

Lectures at PhD Schools

- Mar 2025 Lecturer, **"Tidal Effects in Compact Binaries"**, *Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation*, Galileo Galilei Institute for Theoretical Physics, Florence (IT).
- Dec 2024 Lecturer, **"Black Hole Perturbation Theory"**, *I Gravitass School on Fields and Gravity*, Federal University of Pará, Salinópolis (BR).
- Feb 2023 Lecturer, **"Astrophysical Black Holes: Theory and Observations"**, *59th Winter School "Gravity: Classical, Quantum, and Phenomenology"*, Wrocław (PL).
- Mar 2019 Tutor, **"Gravitational Waves and Compact Binaries"**, *Theoretical Aspects of Astroparticle Physics, Cosmology and Gravitation*, Galileo Galilei Institute for Theoretical Physics, Florence (IT).
- Sep 2018 Lecturer, **"Compact Objects"**, *9th School of Astrophysics and Gravitation*, Instituto Superior Técnico, Lisbon (PT).
- Aug 2018 Lecturer, **"Exotic Compact Objects and Their Gravitational-Wave Signatures"**, *ICTS Summer School on Gravitational-Wave Astronomy*, ICTS, Bangalore (IN).

Training experience

PhD supervision [13 in total].

- 2026–on Advisor of **Nicole Grillo**, **Luca Nagni** and **Juhoon Son**, *Gran Sasso Science Institute*.
- 2025–on Advisor of **Akshita Mittal**, *Gran Sasso Science Institute*.
- 2024–on Advisor of **Laura Pezzella**, *Gran Sasso Science Institute*.
- 2024–on Co-advisor of **Manuel Oliveira Mariano**, *University of Aveiro*.
- 2023–2026 Advisor of **Sara Gliorio**, *Gran Sasso Science Institute*.
- 2022–2025 Co-advisor of **Anjali Kugarajh**, *Gran Sasso Science Institute*.
- 2020–2023 Advisor of **Susanna Barsanti**, *Sapienza University of Rome*.

- 2019–2023 Co-advisor of **Massimo Vaglio** and **Andrea Sabbatucci**, *Sapienza University of Rome*.
 2018–2022 Co-advisor of **Gabriel Piovano**, *Sapienza University of Rome*.
 2017–2020 Co-advisor of **Margherita Fasano**, *Sapienza University of Rome*.

Master's [19 in total] and Bachelor's [6 in total] supervision.

- 2025–2026 Co-advisor of **Giorgio Monti [Master]**, **University of Milano-Bicocca**; advisor of **Giovanni Benetti [Master]**, **University of Padua**.
 2025 Co-advisor of **Serafino Bruscia [Master]**, **University of Pisa**.
 2025 Advisor of **Albert Radulea [post-Bachelor internship]**, *Institut Polytechnique de Paris*.
 2024 Advisor of **Chiara Anselmo*[Master]** and **Niccolò D'Onofrio [Master]**, **Sapienza University of Rome**; Co-advisor of **Elisabeth Keppler*[Master]**, **University of Trieste**.
 2023–2024 Co-advisor of **Alessandro Agapito [Master]**, *Sapienza University of Rome*.
 2023 Advisor of **Laura Pezzella*[Master]**, *Sapienza University of Rome*.
 2022–2023 Advisor of **Enzo Figueiredo [post-Master internship]**, *École Normale Supérieure de Lyon*.
 2021–2022 Advisor of **Lavinia Paiella*[Master]**; Co-advisor of **Matteo Della Rocca [Master]** and **Adriano Frattale Mascioli [Master]**, *Sapienza University of Rome*.
 2020–2021 Advisor of **Viola De Renzis [Master]**; Co-advisor of **Tommaso Garbin [Master]**; advisor of **Alessandro Buchicchio [Bachelor]**, *Sapienza University of Rome*.
 2020 Advisor of **Susanna Barsanti*[Master]**, **Giulio Danella [Master]**; advisor of **Francesca Attadio [Bachelor]**, *Sapienza University of Rome*.
 2019–2020 Advisor of **Irene Fabbri [Master]**; Co-advisor of **Vania Vellucci [Master]**, *Sapienza University of Rome*.
 2019 Advisor of **Simone D'Onofrio [Bachelor]**, **Andrea Falzetti [Bachelor]**, and **Danilo Di Ludovico [Bachelor]**, *Sapienza University of Rome*.
 2017–2018 Co-advisor of **Gonçalo Castro [Master]**, **Instituto Superior Técnico**; Co-advisor of **Giuseppe Gnocchi [Master]**, *Sapienza University of Rome*.
 2017 Co-advisor of **Verena Krall [Bachelor]**, *Eberhard Karls University of Tübingen*.

Institutional duties

- 2022–on **PhD Internal Examiner (Astroparticle Physics)**, *Gran Sasso Science Institute (GSSI)*.
 2021–2023 **Co-organizer, L'Aquila Joint Astroparticle Colloquia**, *Gran Sasso Science Institute (GSSI)*, ([link](#)).
 2021 & 2022 **Co-organizer, Astroparticle Physics Science Fair**, *Gran Sasso Science Institute (GSSI)*.
 2021, 2022, 2025 **PhD Hiring Committee Member**, *Gran Sasso Science Institute (GSSI)*.
 2018–2020 **Co-organizer, Cattedra Fermi Lecture Series**, *Sapienza University of Rome*.

Conference, Workshop, and School Organization

- 5-8 Jul 2027 **2027 LISA Consortium meeting**, Gran Sasso Science Institute, L'Aquila (IT), Organizing Committee.
 9-11 Feb 2027 **TEONGRAV Workshop 2027**, Gran Sasso Science Institute, L'Aquila (IT), Organizing Committee.

^{1*} Recipient of the "Milla Baldo Ceolin" National Prize (INFN) for best Master's thesis in theoretical physics.

- 9-20 Nov 2026 **Theory Meets Experiments School**, Galileo Galilei Institute, Florence (IT), Scientific Committee.
- 28 Sep–2 Oct 2026 **LGWA Meeting 2026**, University of Liège, Liège (BE), Scientific Committee.
- 4 Dec 2025 **Ten Years of Gravitational Waves – An Event at Columbia’s Italian Academy**, Italian Academy for Advanced Studies, Columbia University (USA), Organizing and Scientific Committee.
- 23–25 Sep 2025 **Crossroads in Strong Gravity: Challenges & Future Directions**, Catania (IT), Organizing and Scientific Committee.
- 15–19 Sep 2025 **LGWA Workshop**, University of Camerino, Camerino (IT), Scientific Committee.
- 1–3 Sep 2025 **DeciHz GW Observations on the Moon and Beyond**, Johns Hopkins University, Baltimore (USA), Organizing and Scientific Committee.
- 25–30 May 2025 **Gravitational-Wave Probes of Black Hole Environments**, IFPU, Trieste (IT), Scientific Committee.
- 7–9 May 2025 **Theoretical Horizons in Unraveling Relativity, Astrophysics, and Mergers**, Gran Sasso Science Institute, L’Aquila (IT), Organizing Committee.
- 16–18 Dec 2024 **1st GSSI–JHU MAECI Meeting on Gravitational-Wave Astrophysics**, Assergi, L’Aquila (IT), Organizing Committee.
- 16–20 Oct 2024 **1st TEONGRAV International Workshop on Theory of Gravitational Waves**, Sapienza University of Rome, Rome (IT), Scientific Committee.
- 7–10 Oct 2024 **Lunar Gravitational Wave Antenna Workshop**, Castel Gandolfo (IT), Scientific Committee.
- 1–3 Sep 2024 **Kostas Fest**, Ermoupolis, Syros (EL), Scientific Committee.
- 17–19 Apr 2024 **Linking Advances in Theoretical Astrophysics and Relativity to Observations**, University of Milano-Bicocca, Milano (IT), Organizing Committee.
- 9–13 Oct 2023 **Lunar Gravitational Wave Antenna Workshop**, Catania (IT), Scientific Committee.
- 4–6 Oct 2023 **IV Gravi-Gamma-Nu Workshop**, Gran Sasso Science Institute, L’Aquila (IT), Organizing Committee.
- 20–22 Sep 2023 **Asymmetric Binaries Meet Fundamental Astrophysics**, Gran Sasso Science Institute, L’Aquila (IT), Organizing and Scientific Committee.
- 17–20 Apr 2023 **Lunar Gravitational-Wave Detection**, ICTS, Bangalore (IN), Scientific Committee.
- 5–7 Oct 2022 **The Complexity of the Cosmos: An Interdisciplinary Symposium in Cosmology, Astrophysics, and Statistical Mechanics**, Gran Sasso Science Institute, L’Aquila (IT), Organizing Committee.
- 15–17 Jun 2022 **EuCAPT Workshop: Gravitational-Wave Probes of Black Hole Environments**, Sapienza University of Rome, Rome (IT), Organizing and Scientific Committee.
- 16–19 May 2022 **PHAROS Global Meeting**, Sapienza University of Rome, Rome (IT), Organizing Committee.
- 29 Nov–3 Dec 2021 **IAU Symposium 363**, Gran Sasso Science Institute, L’Aquila (IT), Organizing Committee.
- 10–11 Oct 2019 **Neutron Star Theory Workshop**, Madrid (ES), Scientific Committee.
- 28–30 Aug 2019 **Lost in Gravity**, Saint-Flour (FR), Organizing and Scientific Committee.

- 17–18 Dec 2018 **XI Black Holes Workshop**, Instituto Superior Técnico, Lisbon (PT), Organizing Committee.
- 12–13 Apr 2018 **Neutron Stars in Lisbon**, Instituto Superior Técnico, Lisbon (PT), Organizing and Scientific Committee.

Commissions of Trust

Referee Activity

Living Reviews in Relativity; Czech Academy of Sciences; Dutch Research Council; Physical Review Letters; Physical Review D; The Astrophysical Journal; Journal of Cosmology and Astroparticle Physics; General Relativity and Gravitation; International Journal of Modern Physics; European Physical Journal C; Physics Letters B.

External Examiner

- Nov 2026 **Ciro De Simone**, PhD, University of Naples Federico II (IT).
- Oct 2026 **Simone D'Onofrio**, PhD, Universitat Autònoma de Barcelona (ES).
- Sep 2025 **Clemente Smarra**, PhD, SISSA (IT).
- Mar 2025 **Daniele Pica**, PhD, University of Perugia (IT) & Niels Bohr Institute (DK).
- Apr 2024 **Arianna Foschi**, PhD, Instituto Superior Técnico (PT).
- Jul 2023 **Francisco Duque**, PhD, Instituto Superior Técnico (PT).
- Mar 2023 **Stefano Bondani**, PhD, University of Insubria (IT).
- Oct 2022 **Andrea Miani**, PhD, University of Trento (IT), and J. G. Lara Delgado, PhD, SISSA (IT).

Editorial Activity

- 2024 **Co-editor of the book**, *"Nuclear Theory in the Age of Multimessenger Astronomy"*, CRC Press, ISBN 9781003306580, 2024.

Conferences & Seminars

Chairing

- Jul 2026 **EMRI Search and Inference within the LISA Global Fit - Part II**, National University of Singapore, Singapore (SG).
Panel moderator: "Modelling and Measuring Beyond-Vacuum-GR Effects" (organized and led discussion)
- Oct 2024 **GravityShapePisa**, University of Pisa, Pisa (IT).
Panel moderator: "Fundamental Physics" (co-chair with E. Maggio; organized and led discussion)
- Jul 2024 **Physics & Astrophysics at the Extreme (PAX)**, King's College London, London (GB).
Session chair: "Multimessenger Observations" (co-chair with Marta Zsuzska and Nikhil Sarin; led discussion)
- Dec 2023 **TEONGRAV Meeting**, University of Naples Federico II, Naples (IT).
Session chair: "Tests of gravity, numerical simulations, and observable measurements"
- Aug 2023 **Fundamental Physics Working Group Meeting**, Niels Bohr Institute, Copenhagen (DK).
Panel moderator: "Astro & waveform systematics" (co-chair with D. Shoemaker; organized and led discussion)

- Jun 2023 **EMRIs as Probes of Fundamental Physics**, SISSA, Trieste (IT).
Panel moderator: "Environmental effects on binary black holes" (co-chair with P. Pani; organized and led discussion)
- Feb 2023 **Lunar Gravitational-Wave Antenna - white paper kick-off**, Gran Sasso Science Institute, L'Aquila (IT).
Session chair: "GW science and multi-messenger astronomy" (co-chair with R. Serafinelli; led discussion)
- Nov 2021 **Neutron Star Astrophysics at the Crossroads: Magnetars and the Multimessenger Revolution (IAU Symposium 363)**, Gran Sasso Science Institute, L'Aquila (IT) (online).
Session chair: "NS through GW observations: modelling, search and implications in fundamental physics and astrophysics"
- Nov 2021 **Neutron Star Astrophysics at the Crossroads: Magnetars and the Multimessenger Revolution (IAU Symposium 363)**, Gran Sasso Science Institute, L'Aquila (IT) (online).
Poster session chair
- May 2019 **Physics & Astrophysics at the Extreme (PAX)**, EGO, Pisa (IT).
Session chair: "Compact Binary Coalescences" (co-chair with S. Vitale, J. Veitch and T. Dent)
- Aug 2017 **eXtreme Matter meets eXtreme Gravity**, Montana State University, Bozeman (USA).
Session chair: "New directions and exotic scenarios" (co-chair with Tanja Hinderer)

Invited seminars and talks

- Jun 2027 **[174] TBA**, Invited conference talk: *Synergies for Black-Hole Binaries: from perturbative techniques to strong-field gravity*, Galileo Galilei Institute, Florence (IT).
- Sep 2026 **[173] Environmental effects and where to find them**, Invited conference talk: *13th Aegean Summer School*, Fri, Kasos (EL).
- Aug 2026 **[172] Matters of gravity: the interplay between environmental and beyond-GR effects in asymmetric binaries**, Invited conference talk: *New Horizons for Psi*, Beijing (CHN).
- Jun 2026 **[171] How does beyond vacuum General Relativity sound?**, Invited conference talk: *Symphony of Spacetime: Celebrating the Discovery of Black Hole Quasinormal Modes by C. V. Vishveshwara*, Raman Research Institute, Bangalore (IN).
- Feb 2026 **[170] Challenging the vacuum binary paradigm with gravitational wave observations**, Invited workshop talk: *Gravity Workshop: Mergers and Beyond*, University of Tübingen, Tübingen (DE).
- Feb 2026 **[169] Probing Vacuum General Relativity: Where to Look and What to Seek?**, Invited seminar: *Gr@v Seminar*, University of Aveiro, Aveiro (PT).
- Dec 2025 **[168] Dancing in the Dark (Matter): How Astrophysical Environments Shape Black Hole Encounters**, Invited seminar, Italian Academy of Advanced Studies, Columbia University, New York (USA).
- Dec 2025 **[167] Scale Hierarchies in Gravitational-Wave Binaries: A New Path for Probing Fundamental Physics**, Invited seminar: *Theory Seminar*, Columbia University, New York (USA).
- Oct 2025 **[166] Black hole spectroscopy beyond Kerr with ParSpec**, Invited working group talk: *Beyond GR & Ringdown Group – LISA Fundamental Physics Working Group*, (online).
- Oct 2025 **[165] Scale Asymmetries in Gravitational Wave Sources and Their Physics Payoff**, Invited conference talk: *10 Years of Gravitational Wave Discovery: Advancing the Frontiers of Gravitational Physics*, Indian Institute of Technology Gandhinagar, Gandhinagar (IN).
- Jul 2025 **[164] Environmental effects on black hole physics**, Invited conference talk: *GR24 & Amaldi 16*, Glasgow (GB).

- Jun 2025 **[163] Asymmetric binaries: a tale of two sides**, Invited conference talk: *Black Holes, Neutron Stars, and Gravitational Waves @ Black Sea*, Varna (BG).
- May 2025 **[162] Matters of Relativity**, Invited workshop talk: *Gravitational Wave Probes of Black Hole Environments*, IFPU, Trieste (IT).
- Mar 2025 **[161] Probing Fundamental Physics Through Asymmetric Binary Mergers**, Invited seminar, University of Perugia, Perugia (IT).
- Mar 2025 **[160] Gravitational wave observations of black hole mimickers: where do we look, and what do we look for**, Invited conference talk: *Black Hole Mimickers: From Theory to Observation*, Princeton University, Princeton (USA).
- Oct 2024 **[159] Matter of Love**, Invited colloquium, INAF-Astronomical Observatory of Abruzzo, Teramo (IT).
- Sep 2024 **[158] Black Hole Spectroscopy beyond GR: Theory and Phenomenology**, Invited conference talk: *12th Aegean Summer School*, Apollonia, Sifnos (EL).
- Sep 2024 **[157] Detecting fundamental fields with LISA observations of extreme mass ratio inspirals**, Invited conference talk: *9th Roma International Conference on Astro-Particle Physics*, Frascati (IT).
- Jul 2024 **[156] EMRIs: Beyond-GR, Ringdown and Environmental Effects**, Invited conference talk: *15th LISA Symposium*, Dublin (IE).
- Jul 2024 **[155] Dark matter signatures around dirty black hole binaries**, Invited conference talk: *New Horizons for Psi*, Instituto Superior Técnico, Lisbon (PT).
- Jun 2024 **[154] Fundamental fields, scalar charges and Extreme Mass Ratio Inspirals**, Invited conference talk: *Athens Symposium on Exploring the Universe*, National Hellenic Research Foundation, Athens (EL).
- Mar 2024 **[153] Gravitational wave signatures of asymmetric binaries surrounded by generic dark matter distributions**, Invited journal club talk: *Cosmology Journal Club*, University of Padua, Padua (IT) (online).
- Feb 2024 **[152] How dirty can a black hole be**, Invited workshop talk: *The New Era of Multi-Messenger Astroparticle Physics*, IFPU, Trieste (IT).
- Feb 2024 **[151] A moon window onto deciHz gravitational wave sources**, Invited seminar, Institute of Space and Astronautical Science, Tokyo (JPN).
- Oct 2023 **[150] Black hole spectroscopy beyond Kerr**, Invited conference talk: *GravityShapePisa*, University of Pisa, Pisa (IT).
- Aug 2023 **[149] The relevance of being small: detecting fundamental fields with LISA observations of Extreme Mass Ratio Inspirals**, Invited seminar (webinar), Latin American Webinar on Physics, (online).
- Aug 2023 **[148] Fundamental fields, asymmetric binaries, and LISA observations within a self-force approach**, Invited conference talk: *Gravitational Waves Meet Effective Field Theories*, Benasque (ES).
- Aug 2023 **[147] Astrophysical and Waveform Systematics from the Working Package Perspective**, Invited conference talk: *Fundamental Physics Working Group Meeting*, Niels Bohr Institute, Copenhagen (DK).
- Jun 2023 **[146] Challenging GR with future detectors: the case of 3G ground based interferometers**, Invited conference talk: *Connecting the Dots*, Max Planck Institute for Gravitational Physics, Potsdam (DE).
- Jun 2023 **[145] Black hole spectroscopy beyond Kerr: agnostic vs theory-based tests with 3G interferometers**, Invited conference talk: *Gravitational Universe: Challenges and Opportunities*, SISSA, Trieste (IT).

- May 2023 **[144] Asymmetric binaries as astrophysical laboratories to probe new fundamental fields**, Invited seminar, University of Aveiro, Aveiro (PT).
- May 2023 **[143] Dirty black hole binaries as probe of dark matter halos**, Invited conference talk: *The Dark Matter and Stars: Multi-Messenger Probes of Dark Matter and Modified Gravity*, Instituto Superior Técnico, Lisbon (PT).
- Apr 2023 **[142] Gravitational waves from extreme-mass-ratio systems in astrophysical environments**, Invited seminar, University of Bremen, Bremen (DE) (online).
- Apr 2023 **[141] Gravitational Wave Sources in the deciHz band**, Invited conference talk: *Lunar Gravitational-Wave Detection*, ICTS, Bangalore (IN).
- Mar 2023 **[140] Asymmetric binaries as probe of fundamental astrophysics**, Invited seminar, Max Planck Institute for Gravitational Physics, Potsdam (DE).
- Nov 2022 **[139] The many faces of Extreme Mass Ratio Inspirals**, Invited seminar, University of Pennsylvania, Philadelphia (USA).
- Sep 2022 **[138] Environmental Effects on Gravitational Wave Emission from Extreme Mass Ratio Inspirals**, Invited conference talk: *11th Aegean Summer School*, Ermoupolis, Syros (EL).
- Jun 2022 **[137] The relevance of being small: testing gravity with LISA and extreme mass ratio inspirals**, Invited seminar, University of Milano-Bicocca, Milan (IT).
- Jun 2022 **[136] Common Tools for the Einstein Telescope Community**, Invited conference talk: *XII Einstein Telescope Symposium*, Hungarian Academy of Sciences, Budapest (HU).
- Apr 2022 **[135] The Science Case of the Lunar Gravitational Space Wave Antenna**, Invited working group talk: *LGWA WG1 – GW Science and Multi-messenger Astronomy*, (online).
- Feb 2022 **[134] Testing General Relativity with LISA observations of extreme mass ratio inspirals**, Invited seminar, Institute of Cosmology and Gravitation, Portsmouth (GB) (online).
- Sep 2021 **[133] Detecting new fundamental fields with LISA**, Invited seminar, Instituto Superior Técnico, Lisbon (PT).
- Mar 2021 **[132] Scalar charges, extreme mass ratio inspirals and LISA**, Invited seminar, LUTH – Observatoire de Paris, Paris (FR) (online).
- Feb 2021 **[131] EMRIs in beyond-GR theories with scalar fields**, Invited working group talk: *LISA EMRI Work Package Meeting*, (online).
- Feb 2021 **[130] Constraining the neutron star equation of state with electromagnetic observations and gravitational waves**, Invited seminar, Brera Astronomical Observatory, Merate (IT) (online).
- Dec 2020 **[129] Extreme mass ratio inspirals as probes of scalar fields**, Invited seminar, University of Tübingen, Tübingen (DE) (online).
- Nov 2020 **[128] Extreme mass ratio inspirals as probes of scalar fields**, Invited seminar, SISSA, Trieste (IT) (online).
- Aug 2020 **[127] Neutron stars in and beyond GR**, Invited conference talk: *Probing Effective Theories of Gravity in Strong Fields and Cosmology*, Kavli Institute for Theoretical Physics, Santa Barbara (USA) (online).
- Jun 2020 **[126] NS EoS using multiband independent measurements of radii and tidal deformabilities**, Invited conference talk: *Science case study and scientific simulations for eXTP*, (online).
- Jun 2020 **[125] Testing alternative theories of gravity with QPO and Fe line**, Invited conference talk: *Science case study and scientific simulations for eXTP*, (online).
- Dec 2019 **[124] The Power of Love**, Invited seminar, University of Ferrara, Ferrara (IT).

- Jun 2019 **[I23] Testing the black hole paradigm with gravitational wave binary events**, Invited seminar, Max Planck Institute for Gravitational Physics, Potsdam (DE).
- Feb 2019 **[I22] A heated Love for ECO**, Invited seminar, Johns Hopkins University, Baltimore (USA).
- Jun 2018 **[I21] Gravitational wave phenomenology of exotic compact objects**, Invited seminar, Albert Einstein Institute, Hannover (DE).
- Jun 2018 **[I20] Beyond black holes (exotic compact objects)**, Invited conference talk: *Numerical Relativity Beyond General Relativity*, Benasque (ES).
- Mar 2018 **[I19] Solving the relativistic inverse stellar problem through gravitational-wave observations of binary neutron stars**, Invited conference talk: *Problemi Attuali di Fisica Teorica 2018*, Vietri sul Mare (IT).
- Mar 2018 **[I18] Staring at the black hole horizon with gravitational wave observations**, Invited seminar, Kindai University, Osaka (JPN).
- Jan 2018 **[I17] Gravitational waves as a probe of the equation of state of nuclear matter**, Invited seminar, Sapienza University of Rome, Rome (IT).
- Jan 2018 **[I16] Observational signatures of exotic compact objects**, Invited seminar, Instituto de Astrofísica e Ciências do Espaço, Lisbon (PT).
- Dec 2017 **[I15] Challenging the black hole scenario with gravitational wave observations**, Invited seminar, University of Nottingham, Nottingham (GB).
- Oct 2017 **[I14] The strong gravity regime in the gravitational wave era**, Invited seminar, SISSA, Trieste (IT).
- May 2017 **[I13] Strong-field tests of General Relativity in the black hole arena**, Invited seminar, Aristotle University of Thessaloniki, Thessaloniki (EL).
- Mar 2017 **[I12] Signals from the horizon: General Relativity and the strong-field gravity regime**, Invited seminar, Goethe University Frankfurt, Frankfurt (DE).
- Sep 2016 **[I11] Constraining modified theories of gravity using the stochastic gravitational-wave background**, Invited conference talk: *Gravitational Waves in Modified Gravity Theories*, National Technical University of Athens, Athens (EL).
- Aug 2016 **[I10] Fab Four Neutron Stars**, Invited conference talk: *First TUSUR Cosmological Conference*, Tomsk State University of Control and Radioelectronics, Tomsk (RUS).
- May 2016 **[I9] Gravitational waves: a new window onto the high-energy Universe**, Invited working group talk: *Astronomy Group Meeting*, University of Stuttgart, Stuttgart (DE).
- Mar 2016 **[I8] Relativistic tidal effects in non-standard Kerr spacetime**, Invited conference talk: *Problemi Attuali di Fisica Teorica 2016*, Vietri sul Mare (IT).
- Jun 2015 **[I7] Quasi-periodic oscillations of accreting black holes and the strong-field gravity regime**, Invited seminar, Instituto de Ciencias Físicas, UNAM, Mexico City (MEX).
- Dec 2014 **[I6] Gravitational waves and the physics of compact objects: from gamma-ray bursts to I-Love-Q**, Invited seminar, University of Melbourne, Melbourne (AUS) (online).
- Sep 2014 **[I5] Gravitational waves and the physics of compact objects: from gamma-ray bursts to I-Love-Q**, Invited seminar, Center for Relativistic Astrophysics, Georgia Institute of Technology, Atlanta (USA).
- Apr 2014 **[I4] Low-latency search for gravitational waves from BH-NS binaries in coincidence with short gamma-ray bursts**, Invited conference talk: *Problemi Attuali di Fisica Teorica 2014*, Vietri sul Mare (IT).
- Feb 2014 **[I3] Triggering short gamma-ray burst searches with gravitational-wave events**, Invited conference talk: *SFB-Transregio Meeting*, Tübingen (DE).
- Jul 2013 **[I2] Tidal effects in BH-NS and NS-NS binaries**, Invited conference talk: *Stochastic Gravitational Wave Background in the Advanced Detector Era*, Nice (FR).

Mar 2013 **[I1] I-Love-Q (forever)**, Invited seminar, Institute of Cosmology and Gravitation, Portsmouth (GB).

Contributed talks to conferences and workshops

- Sep 2025 **[C33] Einstein Telescope, deciHz Gravitational Wave Observations on the Moon and Beyond**, Johns Hopkins University, Baltimore (USA).
- Apr 2025 **[C32] The prejudice of Love**, *Theoretical Horizons in Unraveling Relativity, Astrophysics, and Mergers*, Gran Sasso Science Institute, L'Aquila (IT).
- Apr 2024 **[C31] Love, the easy road**, *Linking Advances in our Understanding of Theoretical Astrophysics and Relativity to Observations*, University of Milano Bicocca, Milano (IT).
- Jul 2022 **[C30] LISA, EMRIs and new fundamental fields in the gravity sector**, *The 14th International LISA Symposium*, (online).
- Dec 2021 **[C29] Extreme mass ratio inspirals as LISA probes of fundamental fields**, *Rencontre commune des Groupes de travail Formes d'onde et Tests de la relativité générale et théories alternatives du GdR Ondes Gravitationnelles*, Institut de Physique Théorique (IPhT), Paris (FR).
- May 2021 **[C28] Ranking the Love for the neutron star equation of state with third-generation detectors**, *GWADW2021 Gravitational Wave Advanced Detector Workshop*, (online).
- Oct 2020 **[C27] Parametrizing GR deviations within normal modes of rotating black holes with ParSpec**, *TeonGrav meeting*, (online).
- Sep 2020 **[C26] ParSpec: a data-analysis framework for black-hole spectroscopy with multiple events**, *The 13th International LISA Symposium*, (online).
- Jan 2020 **[C25] A parametrized ringdown approach for black-hole spectroscopy of spinning black holes**, *Gravitational Waves, Black Holes and Fundamental Physics*, SISSA, Trieste (IT), [Poster].
- Oct 2019 **[C24] Quasi-periodic oscillations as probe of the neutron star equation of state**, *Future of X-ray Timing*, Amsterdam (NL).
- Jul 2019 **[C23] Gravitational waves in massive gravity: waveforms from extreme-mass-ratio mergers**, *22nd International Conference on General Relativity and Gravitation*, Valencia (ES).
- Feb 2019 **[C22] From multi-band observations to the nuclear matter equation of state**, *EPS Conference on Gravitation 2019*, Sapienza University of Rome, Rome (IT).
- Jan 2019 **[C21] Gravitational waves in massive gravity theories: constraints from extreme-mass-ratio mergers**, *Gravitational Waves, Black Holes and Fundamental Physics*, Athens (EL).
- Sep 2018 **[C20] Exotic compact objects and their gravitational-wave signatures**, *TeonGrav Meeting 2018*, Sapienza University of Rome, Rome (IT).
- Jan 2018 **[C19] To be a black hole or not to be a black hole**, *Gravity@Malta*, University of Malta, Valletta (MT).
- Dec 2017 **[C18] Gravitational wave signatures of black hole mimickers**, *X Black Holes Workshop*, Aveiro University, Aveiro (PT).
- Nov 2017 **[C17] Exotic Compact Objects in the Strong Field Regime**, *Multidisciplinary Centre for Astrophysics (CENTRA)*, Lisbon (PT).
- Jun 2017 **[C16] A Montecarlo approach for the relativistic inverse stellar problem**, *Bridging Nuclear and Gravitational Physics: the Dense Matter Equation of State*, ECT, Trento (IT).
- Feb 2017 **[C15] Testing strong-field gravity with quasi periodic oscillations**, *High Throughput X-ray Astronomy in the eXTP Era conference*, Rome (IT).

- Sep 2016 **[C14] Binary BH gravitational wave background as probe of alternative theories of gravity**, *NEB-17 Recent Developments in Gravity*, Mykonos (EL).
- Oct 2015 **[C13] From macro to micro: universal properties of neutron stars**, *The Modern Physics of Compact Stars and Relativistic Gravity 2015*, Yerevan State University, Yerevan (AM).
- Jul 2015 **[C12] Interacting vacuum-energy in a Post-Friedmann expanding Universe**, *14th Marcel Grossman Congress 2015*, Sapienza University of Rome, Rome (IT).
- Jul 2015 **[C11] Testing the strong field gravity regime with QPOs observations**, *14th Marcel Grossman Congress 2015*, Sapienza University of Rome, Rome (IT).
- Jul 2015 **[C10] Slowly-rotating black hole solutions in Horndeski theory**, *IVth NRHEP Network Meeting*, Sapienza University of Rome, Rome (IT).
- Apr 2015 **[C9] Testing the strong field gravity regime with QPOs astrophysical observations**, *Center for Relativistic Astrophysics weekly meeting*, Atlanta (USA).
- Nov 2014 **[C8] Testing Gravity with QPOs of Black holes: the case of Einstein-Dilaton-Gauss-Bonnet theory**, *5th LOFT Consortium meeting*, Rome (IT).
- Apr 2014 **[C7] Coincidence searches of gravitational waves and short gamma-ray bursts**, *San Cugat Forum on Astrophysics*, San Cugat, Barcelona (ES).
- Feb 2014 **[C6] Universal relations for Neutron Stars: Gravitational Waves and Short Gamma-ray Bursts**, *TeonGrav Meeting 2014*, Sapienza University of Rome, Rome (IT).
- Oct 2013 **[C5] Black hole-neutron star mergers**, *Einstein Telescope Meeting 2013*, Albert Einstein Institute, Hannover (GE).
- Jul 2013 **[C4] I Love Q forever**, *An International Workshop. Neutron stars: Nuclear Physics, Gravitational Waves and Astronomy*, University of Surrey, Guildford (GB).
- Jul 2012 **[C3] Post-Newtonian-Affine tidal interactions**, *13th Marcel Grossman Congress 2012*, Stockholm (SE).
- Apr 2012 **[C2] Post-Newtonian-Affine tidal interactions**, *Problemi Attuali di Fisica Teorica 2012*, Vietri sul Mare (IT).
- Jun 2011 **[C1] Tidal interactions in compact binary systems**, *5th Nuclear Physics School*, Otranto (IT).

Publications

Regular papers

- gr-qc/
2607.09310 **[R81]** Susanna Barsanti, Ollie Burke, **Andrea Maselli**, Thomas Sotirou, Andrew Spiers, Niels Warburton, *"The significance of first post-adiabatic contributions for scalar charge measurements with extreme-mass-ratio inspirals"*, 2026.
- gr-qc/
2606.18341 **[R80]** Albert Radulea, Marcelo Rubio, Konstantinos Kritos, **Andrea Maselli**, *"Orbital evolution of asymmetric binaries within accreting environments"*, Phys. Rev. D, 2026.
- gr-qc/
2605.02633 **[R79]** Simone D'Onofrio, Sayak Datta, **Andrea Maselli**, *"Axial tidal Love numbers of black holes in matter environments"*, Phys. Rev. D, 2026.
- gr-qc/
2603.10116 **[R78]** Sara Gliorio, Matteo Della Rocca, Susanna Barsanti, Leonardo Gualtieri, **Andrea Maselli**, Thomas Sotirou, *"Adiabatic evolution of asymmetric binaries on generic orbits with new fundamental fields I: characterization of gravitational wave fluxes"*, Phys. Rev. D 114, 064003, 2026.

^{1*} Authors of this paper are all in alphabetical order

^{1†} Authors of this paper are all in alphabetical order after the first one

- gr-qc/
2512.07959 [R77] Matteo Della Rocca, Laura Pezzella, Matteo Emanuele Berti, Leonardo Gualtieri, **Andrea Maselli**, "*Quasinormal ringing of Kerr black holes. III. Excitation coefficients for equatorial inspirals from the innermost stable circular orbit*", Phys. Rev. D 114, 044036, 2026.
- gr-qc/
2507.16904 [R76[†]] Giorgio Nicolini, **Andrea Maselli**, Miguel Zilhao, "*Dancing in the dark: probing Dark Matter through the dynamics of eccentric binary pulsars*", SciPost Phys. 20, 068, 2026.
- gr-qc/
2507.04471 [R75*] Sayak Datta, **Andrea Maselli**, "*A multi-parameter expansion for the evolution of asymmetric binaries in astrophysical environments*", SciPost Phys. 20, 080, 2026.
- gr-qc/
2503.16649 [R74[†]] Sara Glorio, Emanuele Berti, **Andrea Maselli**, Nicholas Speeney, "*Extreme mass ratio inspirals in dark matter halos: Dynamics and distinguishability of halo models*", Phys. Rev. D 112, 124050, 2025.
- gr-qc/
2503.10746 [R73] Valerio De Luca, Loris Del Grosso, Francesco Iacovelli, **Andrea Maselli**, Emanuele Berti, "*Tainted Love: systematic biases from ignoring environmental tidal effects in gravitational wave observations*", Phys. Rev. D 111, 124046, 2025.
- gr-qc/
2502.20137 [R72] Anjali Abirami Kugarajh, Marisol Traforetti, **Andrea Maselli**, Sabino Matarrese, Angelo Ricciardone, "*Scalar-induced Gravitational Waves in Modified Gravity*", Journal of Cosmology and Astroparticle Physics 07, 022, 2025.
- gr-qc/
2412.18651 [R71] Laura Pezzella, Kyriakos Destounis, **Andrea Maselli**, Vitor Cardoso, "*Redshift: quasinormal modes of black holes embedded in generic dark matter halos*", Phys. Rev. D 111, 064026, 2025.
- gr-qc/
2412.15131 [R70] Matteo Della Rocca, George Antoniou, Leonardo Gualtieri, **Andrea Maselli**, "*Probing time-dependent scalar wigs with extreme mass ratio inspirals*", Submitted to Phys. Rev. D, 2024.
- gr-qc/
2406.07607 [R69] Lorenzo Speri, Susanna Barsanti, **Andrea Maselli**, Thomas Sotiriou, Niels Warburton, Maarten van de Meent, Alvin Chua, Ollie Burke, Jonathan Gair, "*Probing fundamental physics with Extreme Mass Ratio Inspirals: a full Bayesian inference for scalar charge*", Phys. Rev. D 113, 023036, 2026.
- gr-qc/
2404.19012 [R68] Giulia Ventagli, Pedro Fernandes, **Andrea Maselli**, Antonio Padilla, Thomas Sotiriou, "*Neutron stars and the cosmological constant problem*", Phys. Rev. D 111, 024001, 2025.
- gr-qc/
2404.02479 [R67] Georgios Antoniou, Caio Macedo, **Andrea Maselli**, Thomas Sotiriou, "*Axial perturbations of hairy black holes in generalised scalar-tensor theories*", Phys. Rev. D 110, 024065, 2024.
- gr-qc/
2403.09767 [R66[†]] Sophia Yi, Adrien Kuntz, Enrico Barausse, Emanuele Berti, Mark Ho-YeGB Cheung, Kostantinos Kritos, **Andrea Maselli**, "*Nonlinear quasinormal mode detectability with next-generation gravitational wave detectors*", Phys. Rev. D 109, 124029, 2024.
- gr-qc/
2402.08883 [R65] Nicholas Loutrel, Sajal Mukherjee, **Andrea Maselli**, Paolo Pani, "*Analytical model of precessing binaries using post-Newtonian theory in the extreme mass-ratio limit I: General Formalism*", Phys. Rev. D 110, 024006, 2024.
- gr-qc/
2401.09542 [R64[†]] Matteo Della Rocca, Susanna Barsanti, Leonardo Gualtieri, **Andrea Maselli**, "*Extreme mass-ratio inspirals as probes of scalar fields: inclined circular orbits around Kerr black holes*", Phys. Rev. D 109, 104079, 2024.
- gr-qc/
2401.00932 [R63[†]] Nicholas Speeney, Emanuele Berti, Vitor Cardoso, **Andrea Maselli**, "*Black holes surrounded by generic matter distributions: polar perturbations and energy flux*", Phys. Rev. D 109, 084068, 2024.
- gr-qc/
2311.14803 [R62] **Andrea Maselli**, Sophia Yi, Lorenzo Pierini, Vania Vellucci, Luca Reali, Leonardo Gualtieri, Emanuele Berti, "*Black hole spectroscopy beyond Kerr: agnostic and theory-based tests with next-generation interferometers*", Phys. Rev. D 109, 064060, 2024.

- gr-qc/
2310.02315 [R61†] Andrew Spiers, **Andrea Maselli**, Thomas Sotiriou, "*Measuring scalar charge with compact binaries: High accuracy modelling with self-force*", Phys. Rev. D 109, 064022, 2024.
- gr-qc/
2309.17404 [R60] Nicholas Loutrel, Richard Brito, **Andrea Maselli**, Paolo Pani, "*Relevance of Precession for Tests of the Black Hole No Hair Theorems*", Phys. Rev. D 110, 044003, 2024.
- gr-qc/
2307.01264 [R59] Luca Reali, **Andrea Maselli**, Emanuele Berti, "*The impact of compact binary confusion noise on tests of fundamental physics in next-generation gravitational-wave detectors*", Submitted to Class. Quant. Grav., 2023.
- gr-qc/
2304.01252 [R58*] Vitor Cardoso, Francisco Duque, **Andrea Maselli**, David Pereñíguez, "*Constraints on massive gravity from dipolar mode excitations*", Phys. Rev. D 108, 124003, 2023.
- gr-qc/
2303.08183 [R57] Enzo Figueiredo **Andrea Maselli**, Vitor Carodso, "*Black holes surrounded by generic dark matter profiles: appearance and gravitational-wave emission*", Phys. Rev. D 107, 104033, 2023.
- gr-qc/
2302.13954 [R56] Massimo Vaglio, Costantino Pacilio, **Andrea Maselli**, Paolo Pani, "*Bayesian parameter estimation on boson-star binary signals with a coherent inspiral template and spin-dependent quadrupolar corrections*", Phys. Rev. D 108, 023021, 2023.
- gr-qc/
2212.03888 [R55] Susanna Barsanti, **Andrea Maselli**, Thomas Sotiriou, Leonardo Gualtieri, "*Extreme Mass Ratio Inspirals a probes of massive scalar fields*", Phys. Rev. Lett. 131, 051401, 2023.
- gr-qc/
2212.03343 [R54†] Valerio De Luca, **Andrea Maselli**, Paolo Pani, "*Modeling frequency-dependent tidal deformability for environmental black hole mergers*", Phys. Rev. D 107, 044058, 2023.
- gr-qc/
2210.01133 [R53*] Vitor Cardoso, Kyriakos Destounis, Francisco Duque, Rodrigo Macedo, **Andrea Maselli**, "*Gravitational waves from extreme-mass-ratio systems in astrophysical environments*", Phys. Rev. Lett. 129, 241103, 2022.
- gr-qc/
2207.07452 [R52†] Gabriel Piovano, **Andrea Maselli**, Paolo Pani, "*Constraining the tidal deformability of supermassive objects with extreme mass ratio inspirals and semi-analytical, frequency-domain waveforms*", Phys. Rev. D 107, 024021, 2023.
- astro-ph/
2206.11286 [R51†] Andrea Sabbatucci, Omar Benhar, **Andrea Maselli**, Costantino Pacilio, "*Sensitivity of Neutron Star Observations to Three-nucleon Forces*", Phys. Rev. D 106, 083010, 2022.
- gr-qc/
2204.12510 [R50†] Goncalo Castro, Leonardo Gualtieri, **Andrea Maselli**, Paolo Pani, "*Impact and detectability of spin-tidal couplings in neutron star inspirals*", Phys. Rev. D 106, 020411, 2022.
- gr-qc/
2203.07442 [R49] Massimo Vaglio, Costantino Pacilio, **Andrea Maselli**, Paolo Pani, "*The multipolar structure of rotating boson stars*", Phys. Rev. D 105, 124020, 2022.
- gr-qc/
2203.05003 [R48†] Susanna Barsanti, Nicola Franchini, Leonardo Gualtieri, **Andrea Maselli**, Thomas Sotiriou, "*Extreme mass-ratio inspirals as probes of scalar fields: eccentric equatorial orbits around Kerr black holes*", Phys. Rev. D 106, 044029, 2022.
- gr-qc/
2203.01725 [R47] Nicholas Loutrel, Richard Brito, **Andrea Maselli**, Paolo Pani, "*Inspiralling compact objects with generic deformations*", Phys. Rev. D 105, 124050, 2022.
- Editor Suggestion on PRD
- astrop-ph/
2109.02170 [R46*] Susanna Barsanti, Valerio De Luca, **Andrea Maselli**, Paolo Pani, "*Detecting Subsolar-Mass Primordial Black Holes in Extreme Mass-Ratio Inspirals with LISA and Einstein Telescope*", Phys. Rev. Lett. 11, 111104, 2022.
- gr-qc/
2109.00005 [R45*] Vitor Cardoso, Kyriakos Destounis, Francisco Duque, Rodrigo Macedo, **Andrea Maselli**, "*Black holes in galaxies: environmental impact on gravitational-wave generation and propagation*", Phys. Rev. D (Letter) 105, L061501, 2022.
- gr-qc/
2106.11325 [R44] **Andrea Maselli**, Nicola Franchini, Leonardo Gualtieri, Thomas Sotiriou, Susanna Barsanti, Paolo Pani, "*Detecting fundamental fields with LISA observations of gravitational waves from extreme mass-ratio inspirals*", Nature Astronomy 6, 464-470, 2022.

- gr-qc/ [2105.07083](#) [R43[†]] Gabriel Piovano, Richard Brito, **Andrea Maselli**, Paolo Pani, "Assessing the detectability of the secondary spin in extreme mass-ratio inspirals with fully-relativistic numerical waveforms", Phys. Rev. D 104, 124019, 2021.
- gr-qc/ [2104.10035](#) [R42] Costantino Pacilio, **Andrea Maselli**, Margherita Fasano, Paolo Pani, "Ranking Love Numbers for the Neutron star Equation of State: the Need for Third-Generation Detectors", Phys. Rev. Lett. 128, 101101, 2022.
- astro-ph/ [2010.03581](#) [R41] **Andrea Maselli**, Andrea Sabatucci, Omar Benhar, "Constraining three-nucleon forces with multimessenger data", Phys. Rev. C 103, 065804, 2021.
- gr-qc/ [2007.05264](#) [R40] Costantino Pacilio, Massimo Vaglio, **Andrea Maselli**, Paolo Pani, "Gravitational-wave detectors as particle-physics laboratories: Constraining scalar interactions with boson-star binaries", Phys. Rev. D 102, 083002, 2020.
- astro-ph/ [2005.01726](#) [R39] Margherita Fasano, Kaze Wong, **Andrea Maselli**, Emanuele Berti, Valeria Ferrari, Namgalore Sathyaprakash, "Distinguishing double neutron star from neutron star-black hole binaries with gravitational wave observations", Phys. Rev. D 102, 023025, 2020.
- gr-qc/ [2004.11895](#) [R38] **Andrea Maselli**, Nicola Franchini, Leonardo Gualtieri, Thomas Sotiriou, "Detecting scalar fields with Extreme Mass Ratio Inspirals", Phys. Rev. Lett. 125, 141101, 2020.
- gr-qc/ [2004.02654](#) [R37[†]] Gabriel Piovano, **Andrea Maselli**, Paolo Pani, "Extreme mass ratio inspirals with spinning secondary: a detailed study of equatorial circular motion", Phys. Rev. D 102, 024041, 2020.
- gr-qc/ [2003.08448](#) [R36[†]] Gabriel Piovano, **Andrea Maselli**, Paolo Pani, "Model independent tests of the Kerr bound with extreme mass ratio inspirals", Phys. Lett. B. 811, 135860, 2020.
- gr-qc/ [1910.12893](#) [R35] **Andrea Maselli**, Paolo Pani, Leonardo Gualtieri, Emanuele Berti, "Parametrized ringdown spin expansion coefficients: a data-analysis framework for black-hole spectroscopy with multiple events", Phys. Rev. D 101, 024043, 2020.
- astro-ph/ [1910.00016](#) [R34] **Andrea Maselli**, Stefania Marassi, Marica Branchesi, "White dwarf binaries and deci-Hz gravitational wave observations: from the Hubble constant to the supernova astrophysics", Astronomy & Astrophysics 635, A120, 2020.
- astro-ph/ [1909.05870](#) [R33*] Vitor Cardoso, **Andrea Maselli**, "Constraints on the astrophysical environment of binaries with gravitational-wave observations", Astronomy & Astrophysics 644, A147, 2020.
- gr-qc/ [1906.05155](#) [R32] Ryan McManus, Caio Macedo, Masashi Kimura, Emanuele Berti, **Andrea Maselli**, Vitor Cardoso, "Parametrized black hole quasinormal ringdown. II. Coupled equations and quadratic corrections for nonrotating black holes", Phys. Rev. D 100, 044061, 2019.
- gr-qc/ [1905.13460](#) [R31] Giuseppe Gnocchi, **Andrea Maselli**, Tiziano Abdelsalhin, Giuseppe Giacobbo, Michela Mapelli, "Bounding Alternative Theories of Gravity with Multi-Band GW Observations", Phys. Rev. D 100 064024, 2019.
- astro-ph/ [1905.05769](#) [R30] **Andrea Maselli** Chris Kouvaris, Kostas Kokkotas, "The Photon Spectrum of Asymmetric Dark Stars", International Journal of Modern Physics D, 2150003, 2020.
- astro-ph/ [1905.05616](#) [R29] **Andrea Maselli**, George Pappas, Paolo Pani, Leonardo Gualtieri, Sara Motta, Valeria Ferrari, Luigi Stella, "A new method to constrain neutron star structure from quasi-periodic oscillations", The Astrophysical Journal 899 139, 2020.
- gr-qc/ [1905.03947](#) [R28] Paolo Pani, **Andrea Maselli**, "Love in Extrema Ratio", International Journal of Modern Physics D, 101142, 2019.
- Honorable mention of the Gravity Research Foundation Essay Competition**
- astro-ph/ [1902.05078](#) [R27[†]] Margherita Fasano, Tiziano Abdelsalhin, **Andrea Maselli**, Valeria Ferrari, "Constraining the neutron star equation of state using multi-band independent measurements of radii and tidal deformabilities", Phys. Rev. Lett. 123 141101, 2019.

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1901.01265 [R26*] Vitor Cardoso, Masashi Kimura, **Andrea Maselli**, Emanuele Berti, Caio Macedo, Ryan McManus, "*Parametrized black hole quasinormal ringdown: Decoupled equations for nonrotating black holes*", Phys. Rev. D 99 104077, 2019.
- gr-qc/
1811.03689 [R25] **Andrea Maselli**, Paolo Pani, Vitor Cardoso, Tiziano Abdelsalhin, Leonardo Gualtieri, Valeria Ferrari, "*From micro to macro and back: probing quantum structures at the horizon scale with gravitational waves*", Class. Quant. Grav. 36 167001, 2019.
- gr-qc/
1809.00673 [R24*] Vitor Cardoso, Goncalo Castro, **Andrea Maselli**, "*Gravitational waves in massive gravity theories: waveforms, fluxes and constraints from extreme-mass-ratio mergers*", Phys. Rev. Lett. 121 251103, 2018.
- gr-qc/
1808.08962 [R23*] Vitor Cardoso, Masashi Kimura, **Andrea Maselli**, Leonardo Senatore, "*Black holes in EFT extensions of GR*", Phys. Rev. Lett. 121 251105, 2018.
- gr-qc/
1712.01303 [R22] Tiziano Abdelsalhin, **Andrea Maselli**, Valeria Ferrari, "*Solving the relativistic inverse stellar problem through gravitational waves observation of binary neutron stars*", Phys. Rev. D 97 084014, 2018.
- gr-qc/
1708.02217 [R21] **Andrea Maselli**, Sebastian H. Völkel, Kostas D. Kokkotas, "*Parameter estimation of gravitational wave echoes from exotic compact objects*", Phys. Rev. D 96 064065, 2017.
- astro-ph/
1704.07286 [R20] **Andrea Maselli**, Pantelis Pnigouras, Niklas Grønlund Nielsen, Chris Kouvaris, Kostas Kokkotas, "*Dark stars: gravitational and electromagnetic observables*", Phys. Rev. D 96 023005, 2017.
- gr-qc/
1703.10612 [R19] **Andrea Maselli**, Paolo Pani, Vitor Cardoso, Tiziano Abdelsalhin, Leonardo Gualtieri, Valeria Ferrari, "*Probing Planckian corrections at the horizon scale with LISA binaries*", Phys. Rev. Lett. 120 081101, 2018.
- Editor suggestion in PRL**
- astro-ph/
1703.01472 [R18] **Andrea Maselli**, Paolo Pani, Roberto Cotesta, Leonardo Gualtieri, Luigi Stella, Valeria Ferrari, "*Geodesic models of quasi-periodic-oscillations as probes of quadratic gravity*", The Astrophysical Journal 843 1, 2017.
- gr-qc/
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1806.05195 [C1] L. Barack *et al*, "*Black holes, gravitational waves and fundamental physics: a roadmap*", *Classical and Quantum Gravity* 36 143001, 2018.
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Books & contributions to books

[B1] **Andrea Maselli**, Francesco Pannarale, "*Multimessenger constraints on the neutron star equation of state microphysics*", Chapter contribution for the book "*Nuclear Theory in the Age of Multimessenger Astronomy*", CRC Press, ISBN 9781003306580, 2024.

Proceedings

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2505.09014 [P9] **Andrea Maselli**, "*Gravitational wave observations of black hole mimickers: where do we look, and what do we look for*", Proceeding of the conference "Black hole mimickers: from theory to observation", Princeton (USA), , 2025.
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2311.01911 [P8] Martina Adamo, **Andrea Maselli**, "*Astrophysical black holes*", Proceeding of the Winter School of Theoretical Physics and Third Training School of COST Action CA18108, Proceedings of Science QG-MMSchools 011, 2023.
- [P7] Paolo Pani, Leonardo Gualtieri, **Andrea Maselli**, Valeria Ferrari, "*Recent progress on the tidal deformability of spinning compact objects*", Proceeding of the 14th Marcel Grossman Congress, Rome (IT), World Scientific Publishing, ISBN 978-981-3226-59-3, 2017.
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1602.05997 [P4] Hector O. Silva, **Andrea Maselli**, Masato Minamitsuji, Emanuele Berti, "*Compact objects in Horndeski gravity*", Proceeding of the III Amazonian Symposium on Physics and 5th NRHEP Network Meeting is approaching: Celebrating 100 Years of General Relativity, Belen (BR), 2015, *International Journal of Modern Physics D*, 1641006, 2016.
- [P3] **Andrea Maselli**, "*From macro to micro: universal properties of neutron stars*", Proceeding of the conference: The Modern Physics of Compact Stars 2015, Yerevan (AM), Proceedings of Science MPC2015 014, 2016.

- gr-qc/1302.5249 **[P2] Andrea Maselli**, Leonardo Gualtieri, Francesco Pannarale, Valeria Ferrari, *"On the validity of the adiabatic approximation in compact binary inspirals"*, Proceeding of the 13th Marcel Grossman Meeting, Stockholm (S), World Scientific Publishing, ISBN 978-981-4612-14-2, 2015.
- gr-qc/1405.7201 **[P1] Andrea Maselli**, Valeria Ferrari, *"Coincidence searches of gravitational waves and short gamma-ray bursts"*, Proceeding of the San Cugat Forum on Astrophysics, San Cugat (E), Springer International Publishing, Astrophysics and Space Science Proceedings, 40 75-83, 2014.

Outreach Activities

- Jul 2022 **Podcast Episode**, *"Sarà Scienza o Sarà Fake?"*, Contribution to the podcast "Sidecar" (Gran Sasso Science Institute), interview with Dr. Barbara Gallavotti, ([link](#)).
- Dec 2021 **Outreach Talk**, *"Questione di Spazio-Tempo"*, IIS Celestino Rosatelli, Rieti (IT).
- Oct 2021 **Outreach Talk**, *"Matter of Space-Time"*, Public outreach event "Views on Time", Cusano University, Rome (IT).
- Sep 2021 **Member of the GSSI Outreach Gravity Team**, *SHARPER Night*, European Researchers' Night, L'Aquila (IT), ([link](#)).
- Mar 2021 **Podcast Episode**, *"La solitudine dell'Homo sapiens"*, Contribution to the podcast "Sidecar" (Gran Sasso Science Institute), interview with Prof. Telmo Pievani, ([link](#)).
- Nov 2020 **Outreach Video**, *"Gravitational Waves: A New Window on Gravity"*, European Researchers' Night, ([link](#)).
- Oct 2018 **Outreach Essay**, *"Fertilising Ideas"*, Contribution to the project "The Birth of an Idea", ([link](#)).
- Jul 2018 **Marker of Theoretical Exams, IPhO 2018**, *International Physics Olympiad*, Lisbon (PT), 21–29 July.
- Sep 2012 **Outreach Article**, *Andrea Maselli & Niccolò Loret, "Non si sorpassa in tunnel"*, Review article for the student scientific magazine *Accastampato*, Physics Department, Sapienza University of Rome, Accastampato 9 (2012) ([link](#)).

Miscellaneous

Computer Skills

Programming	Fortran, C, Mathematica
Operating Systems	macOS, Linux, Windows
Scientific Software	Mathematica, Maple, MATLAB, Origin, GRTensor, Gnuplot
Other Software	LaTeX, Microsoft Office

Languages

Italian	Native
English	Fluent

Interests

- Biophysics
- Computer Science
- Travel
- Charity work, including a one-month volunteering experience in Kinshasa (Democratic Republic of the Congo)
- Technology
- Photography
- Martial Arts

Andrea Maselli